

Backend Features

This document provides a comprehensive breakdown of all backend features required for the e-commerce marketplace, organized by priority and functional areas.

Priority Levels

Priority	Meaning	Description
 P0 — Critical	App cannot function without it	Core functionality required for basic operations.
 P1 — High	Required before launch	Features necessary for a complete and secure MVP.
 P2 — Medium	Post-launch enhancements	Important features that can be added after the initial release.
 P3 — Low	Deferred / nice-to-have	Extra features to improve UX or operational efficiency.

1. Auth & Authorization

-  **P0** User registration and login (Email/Password).
-  **P0** JWT generation and validation.
-  **P1** Role-Based Access Control (RBAC) - Admin, Vendor, Customer.
-  **P1** Refresh tokens mechanism.
-  **P2** Social Login (Google, Facebook).
-  **P2** Password reset via email.

2. User Management

-  **P0** Create, read, update user profile.
-  **P1** Multiple address management (CRUD for shipping/billing addresses).
-  **P2** User account deletion/deactivation.

3. Vendor Management

-  **P0** Vendor registration and onboarding.
-  **P1** Vendor approval/rejection flow (by Admin).
-  **P1** Vendor profile and store settings.
-  **P2** Vendor performance rating APIs.

4. Product & Catalog

-  **P0** Product CRUD operations (restricted to Admin/Vendor).
-  **P0** Category and subcategory management.
-  **P1** Product variants (size, color, etc.) and SKUs.
-  **P1** Product reviews and ratings.
-  **P2** Bulk product import/export (CSV).

5. Shopping Cart

-  **P0** Add/Remove/Update items in the cart.
-  **P0** Calculate total price (including taxes and discounts).
-  **P1** Sync cart across devices (persist in DB).
-  **P2** Abandoned cart reminders.

6. Wishlist

-  **P1** Add/Remove products to/from wishlist.
-  **P2** Share wishlist.
-  **P3** Wishlist price drop alerts.

7. Order Management

-  **P0** Order creation and placement.
-  **P0** Order status updates (Pending, Processing, Shipped, Delivered, Cancelled).
-  **P1** Order cancellation by Customer/Vendor.
-  **P1** Order tracking endpoints.
-  **P2** Return and refund requests flow.

8. Inventory

-  **P0** Stock deduction upon order placement.
-  **P0** Stock rollback on order failure/cancellation.
-  **P1** Low stock alerts for vendors.

9. Payment & Billing

-  **P0** Payment gateway integration (Razorpay/Stripe).
-  **P0** Generate order receipts and invoices.
-  **P1** Payment webhooks handling (success, failure).
-  **P2** Vendor payouts calculation and disbursements.

10. Notifications

-  **P1** Email triggers for registration, order placement, order status changes.
-  **P2** In-app notifications (WebSockets).
-  **P3** SMS templates/notifications.

11. Admin Panel

-  **P0** Admin dashboard stats (total revenue, active users, etc.).
-  **P1** Vendor approval management.
-  **P2** User and vendor suspension/banning.
-  **P2** Audit logs for admin actions.
-  **P3** Support tickets overview.

12. Analytics & Reporting

-  **P1** Sales and revenue endpoints (for Recharts integration).
-  **P2** Top-selling products and top vendors reporting.
-  **P3** Conversion rate tracking.

13. Search & Discovery

-  **P0** Basic product search by name/description.
-  **P1** Advanced filters (price range, category, brand, rating).
-  **P1** Pagination and sorting (price high-low, newest first).
-  **P2** Search auto-complete and suggestions.

14. Support & Customer Care

-  **P2** Create, update, and resolve support tickets.
-  **P2** FAQ management.
-  **P3** Live chat integration endpoint setup.

15. Security & Compliance

-  **P0** Secure password hashing (bcrypt).
-  **P0** Rate limiting on login and sensitive endpoints.
-  **P1** Data sanitization and input validation (OWASP-aligned).
-  **P1** CORS and Helmet setup.
-  **P2** Data export/deletion for GDPR compliance.

16. Media & File Uploads

-  **P0** Image upload endpoints (for Products, Profile avatars).
-  **P1** Cloud storage integration (AWS S3, Cloudinary).
-  **P2** Media optimization and resizing on upload.

17. Marketing & Promotions

-  **P2** Promo codes and coupon generation/validation.
-  **P2** Discount rules (e.g., specific category, cart value threshold).
-  **P3** Affiliate marketing link tracking.

Recommended Implementation Phases

Phase 1: Core Foundation (The Need-to-Haves)

Focus heavily on P0 features. Get the essential structure running.

- Auth & Authorization
- User Management
- Product & Catalog (Basic CRUD)
- Security & Compliance basics
- Media & File Uploads (Basic local or direct cloud)

Phase 2: Marketplace Operations (The E-commerce Core)

Enable buyers to buy and sellers to sell.

- Vendor Management
- Shopping Cart
- Order Management
- Inventory
- Payment & Billing

Phase 3: Engagement & Operations (The Polish)

Build trust, improve UX, and support admin functions.

- Admin Panel
- Search & Discovery (Filters, sorting)
- Notifications (Emails)
- Wishlist

Phase 4: Growth & Scale (The Nice-to-Haves)

Advanced features for scaling the business.

- Analytics & Reporting
- Support & Customer Care
- Marketing & Promotions
- Advanced Security (Rate limiting, Audit logs)

Data Models Summary Table

Model Name	Core Attributes	Relationships
User	ID, Name, Email, PasswordHash, Role, Status	1:M Addresses, 1:M Orders
Vendor	ID, UserID, StoreName, ApprovalStatus, TaxInfo	1:1 User, 1:M Products
Address	ID, UserID, Street, City, State, Zip, Country	M:1 User
Category	ID, Name, ParentID, Slug	1:M Products, 1:M Subcategories
Product	ID, VendorID, CategoryID, Title, Desc, Price, Stock	M:1 Vendor, M:1 Category, 1:M Variants, 1:M Reviews
Variant	ID, ProductID, Color, Size, SKU, PriceAdjust	M:1 Product
Cart	ID, UserID, TotalAmount	1:1 User, 1:M CartItems
CartItem	ID, CartID, ProductID, Quantity, Price	M:1 Cart, M:1 Product

Model Name	Core Attributes	Relationships
Order	ID, UserID, Total, Status, PaymentStatus	M:1 User, 1:M OrderItems, 1:1 Payment
OrderItem	ID, OrderID, ProductID, Quantity, Price	M:1 Order, M:1 Product
Payment	ID, OrderID, TransactionID, Amount, Status	1:1 Order
Review	ID, ProductID, UserID, Rating, Comment	M:1 Product, M:1 User
Wishlist	ID, UserID	1:1 User, M:M Products
Ticket	ID, UserID, Subject, Description, Status	M:1 User